

AXIOM PROTOCOL

Validator Economics

Fair contribution rewards for Guardian Core Nodes, Citizen Validators, Mobile Guardians, and Passive Witnesses.

Trust Mathematics, Not Power.

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Reward philosophy

Rewards should reflect contribution, risk, uptime, correctness, decentralization value, and long-term honesty. They should not be proportional only to wealth.

Reward Score = contribution_score + risk_score + uptime_score + correctness_score + decentralization_bonus + reputation_score - penalty_score

Participation layers

- Guardian Core Node: critical consensus layer; produces blocks, executes contracts, signs finality, and accepts slashing risk.
- Citizen Validator: independent full verification layer; verifies blocks and state, monitors validators, and submits fraud proofs.
- Mobile Guardian: light verification layer; verifies headers, Merkle proofs, signatures, and fraud/validity proofs with bounded risk.
- Passive Witness: low-risk signal layer; relays transactions, reports scams, and contributes availability/reputation signals.

Reward pool split

Recommended starting split: Guardian Core Node 60%, Citizen Validator 22%, Mobile Guardian 10%, Passive Witness 5%, Fraud Bounty 3%.

Reward pool table

Layer	Suggested Share	Rationale
Guardian Core Node	60%	Consensus, block production, execution, finality, slashing risk
Citizen Validator	22%	Independent verification, state checks, fraud proofs
Mobile Guardian	10%	Light proof verification, relay, fraud watching
Passive Witness	5%	Scam reports, availability signals, low-risk participation
Fraud Bounty	3%	Special rewards for proven attacks or misconduct

Scoring by layer

Guardian Core: block participation, uptime, correctness, decentralization bonus, and reputation. Citizen

Validator: verification work, uptime, fraud monitoring, data availability checks, and reputation. Mobile

Guardian: useful proof verification, honest participation, relay support, device/account reputation, and anti-sybil bond. Passive Witness: useful reports, app participation, reputation signal quality, and anti-spam quality.

Anti-whale economics

Axiom should reduce the marginal power of concentrated capital. A whale can participate, but the protocol should reward distributed honest operation more than capital concentration.

Effective Stake = $\sqrt{\text{raw_stake}}$

Per-validator cap:

- no validator may receive more than X% of epoch reward pool
- no validator may exceed Y% of effective voting power

Dynamic decentralization penalty:

- larger validators receive lower marginal APR after saturation threshold

Anti-bot economics for mobile participation

- Mobile rewards must have daily caps.
- Rewards must be based on useful verified work, not app open time alone.
- New accounts and devices should start with low reward eligibility.
- Participation reputation should grow slowly.
- Suspicious clusters should be rate-limited or required to post higher bonds.
- Privacy must be preserved; anti-sybil cannot become mandatory surveillance.

Slashing and penalties

Guardian Core Nodes can be slashed for double-signing, invalid blocks, censorship behavior, or finality attacks.

Citizen Validators and Mobile Guardians should not carry catastrophic slashing risk; instead they may lose rewards, reputation, or small bonds for malicious proof spam or false reports.

Year-101 sustainability

After minting ends, validators must remain economically useful. Rewards should come from market demand: L1 fees, gas, Pulse settlement, data availability, AI transaction volume, security services, fraud bounties, and Commons Reserve support.

Disclaimer

This document is a Phase 0 research and design specification. It is not investment advice, legal advice, security assurance, or a promise that the system is feasible without further research, simulations, audits, and testnet validation.